

HUNTERS POINT, MB, Canada

WMO: 711421

Lat: **53.027N**Lon: **100.935W**Elev: **840**StdP: **14.26**Time Zone: **-6.00 (NAC)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-23.0	-33.8	0.8	-26.7	-29.8	1.1	-22.8	31.0	-1.7	28.1	-2.1	7.5	280	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	80.6	68.0	77.5	68.1	75.1	67.6	72.0	76.9	70.4	75.4	68.5	73.5	7.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	115.1	75.5	68.6	108.3	73.9	66.6	101.0	72.1	36.5	77.2	34.9	75.3	33.3	73.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	29.8	26.3	23.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				WB	-33.2	74.9	3.4	2.2	-35.7	76.4	-37.7	77.7	-39.5	78.9	-42.0	80.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	-1.4	5.9	15.6	29.9	43.8	60.5	67.0	65.2	54.8	39.7	22.4	7.3
	DBStd	25.71	14.02	13.75	14.08	9.76	9.54	7.05	4.95	6.13	7.43	8.52	11.04	14.83
	HDD50	7275	1594	1234	1066	605	247	7	0	0	40	329	829	1324
	HDD65	11399	2059	1654	1531	1054	662	168	36	73	310	784	1279	1789
	CDD50	1572	0	0	0	1	54	322	528	471	185	11	0	0
	CDD65	221	0	0	0	0	4	33	100	79	5	0	0	0
	CDH74	837	0	0	0	0	32	163	343	270	27	2	0	0
	CDH80	151	0	0	0	0	10	49	46	43	3	0	0	0
Wind	WSAvg	10.7	9.7	10.4	10.2	9.9	10.1	10.8	9.9	10.9	12.1	12.7	11.6	10.5
Precipitation	PrecAvg	20.0	0.9	0.6	0.7	1.1	1.8	3.1	3.4	2.7	2.3	1.6	1.0	0.8
	PrecMax	27.7	1.6	1.1	1.7	1.9	4.1	5.4	6.0	6.7	4.3	4.7	2.4	1.8
	PrecMin	15.7	0.4	0.2	0.2	0.2	0.4	2.0	2.0	1.1	0.6	0.2	0.4	0.4
	PrecStd	3.0	0.3	0.2	0.4	0.5	0.9	0.8	1.0	1.4	1.0	0.9	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	40.4	48.9	61.2	78.0	86.1	84.8	86.1	77.5	61.6	45.7	43.1
		MCWB	38.3	35.2	42.0	50.3	56.8	66.6	68.5	67.7	62.3	54.0	38.9	36.9
	2%	DB	34.8	34.5	41.5	48.2	66.4	77.8	80.3	79.4	71.5	56.6	40.5	35.6
		MCWB	32.4	32.9	36.7	41.5	56.6	67.8	68.9	69.7	61.3	49.7	36.5	32.5
	5%	DB	26.8	31.1	37.9	43.2	60.8	73.5	77.6	76.6	67.6	53.3	37.8	30.3
		MCWB	25.2	29.5	34.8	38.5	55.0	65.0	68.4	68.8	60.8	48.6	35.2	28.5
	10%	DB	19.0	26.9	35.0	40.2	57.0	70.1	75.2	74.4	64.4	50.4	35.8	26.0
		MCWB	18.3	25.6	33.2	36.9	51.9	63.3	68.2	67.7	58.9	47.1	33.9	24.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	36.2	41.9	48.0	61.6	72.9	74.7	74.1	65.8	55.4	39.7	37.1
		MCDB	42.9	38.7	48.4	60.3	73.7	77.9	78.9	78.3	70.8	61.6	44.1	43.1
	2%	WB	32.5	33.1	37.3	41.5	57.6	69.7	72.4	71.9	63.4	50.8	37.2	32.5
		MCDB	34.2	34.4	40.7	46.7	64.3	75.9	76.9	77.0	69.0	54.9	39.8	35.1
	5%	WB	25.2	29.4	35.1	38.9	54.8	66.6	70.7	70.2	61.4	49.0	35.7	28.7
		MCDB	26.8	31.2	37.6	42.8	60.2	72.1	75.3	75.5	66.7	52.4	37.7	30.4
	10%	WB	18.3	25.7	33.2	37.0	52.0	64.2	68.9	68.4	59.2	47.3	34.1	25.0
		MCDB	18.9	26.9	35.0	40.0	56.7	69.2	73.5	73.6	63.8	50.4	35.7	26.0
Mean Daily Temperature Range	5% DB	MDBR	15.4	17.8	20.3	16.2	13.3	11.7	13.0	12.2	11.4	9.1	10.5	14.0
		MCDBR	23.4	19.3	22.6	19.9	18.4	18.1	17.0	16.2	17.5	13.5	11.9	18.7
		MCWBR	21.0	17.5	18.6	14.1	10.3	10.2	9.9	9.6	10.3	8.9	9.3	16.5
	5% WB	MCDBR	22.9	18.8	21.1	18.4	16.1	13.7	13.5	13.6	14.6	12.2	11.1	18.5
		MCWBR	20.7	17.1	17.7	13.6	9.6	9.0	9.3	9.4	9.9	8.5	8.8	16.3
Clear-Sky Solar Irradiance	taub	0.252	0.257	0.282	0.347	0.357	0.386	0.390	0.373	0.332	0.294	0.255	0.233	
	taud	2.265	2.311	2.383	2.379	2.412	2.363	2.392	2.434	2.531	2.576	2.429	2.306	
	Ebn at Noon	236	276	291	281	283	275	271	270	271	262	239	225	
	Edh at Noon	19	25	29	34	35	37	35	32	26	20	16	15	
All-Sky Solar Radiation	RadAvg	333	699	1177	1566	1749	1773	1842	1511	1070	599	288	227	
	RadStd	28	41	81	129	161	138	104	126	112	79	42	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	+3.04	N/A	-0.68	-0.77	N/A	N/A	N/A	N/A